

NB46: PAH ve CFTR - Reverse-Distribution Egitim Deneyi

SEED=42 | PI_TEST=0.2 | LOO-CV + Robust Bootstrap 80/20 Protokolu

Yoneticici Ozeti

Amac: Literatur arastirmasinda (reports/literature_research_panel_improvements_2026-07-24.md) en yuksek getirili/en dusuk riskli hamle olarak belirlenen reverse-distribution (gercek %60/40 benign/pathogenic resample ile egitim) tekniginin PAH ve CFTR panellerine uygulanmasi. Bu teknik MASTER (NB39: 0.638) ve KANSER'de (NB32: 0.730) kanitlanmis ama PAH ve CFTR'de hic denenmemisti.

Yontem: Her panel icin COMBINED egitim havuzu (diger 3 panelin birlesimi, hedef panel HARIC) olusturuldu. Bu havuzdan 4 senaryo turetildi: R0 (orijinal dagilim, kontrol), R1 (%60 benign/%40 patho), R2 (%70/%30), R3 (%80/%20 agresif). Her senaryoda LGBM ve BalancedBagging modelleri egitildi. Degerlendirme NB21 (PAH)/NB20 (CFTR) ile BIREBIR AYNI protokolde yapildi: hedef panel uzerinde LOO-CV, robust N=50 bootstrap %80/20 threshold secimi, Saerens prior-shift, ve floor-F1 referansi (kendi test havuzunun prevalansindan).

Ana Bulgular

PAH: En iyi reverse-dist senaryosu R1_REVERSE_6040/lgbm ile Boot-F1(prior)=0.5889 elde edildi (referans NB21 sampiyonu P4_COMBINED_BalBag=0.582, delta=+0.0069). Floor-F1=0.9083.

CFTR: En iyi reverse-dist senaryosu R2_REVERSE_7030/lgbm ile Boot-F1(prior)=0.8377 elde edildi (referans NB20 sampiyonu S0c_PriorShift=0.863, delta=-0.0253). Floor-F1=0.8955.

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PAH - Reverse-Distribution Sonuclari

COMBINED havuzu = MASTER + KANSER + CFTR (PAH panelin kendisi egitime girmiyor, LOO-CV ile test ediliyor).

Deney	n_train	benign_frac	LOO-MCC(prior)	Boot-F1(prior)	Boot-CI	Precision	Recall	FLOOR-F1	Floor Gecti mi
R1_REVERSE_6040/lgbm	1538	0.6	0.3767	0.5889	[0.429-0.706]	0.9643	0.6156	0.9083	HAYIR
REF_P4_COMBINED_BalBag_NB21	-	-	-	0.5820		-	-	0.9083	HAYIR
R1_REVERSE_6040/balbag	1538	0.6	0.436	0.5531	[0.450-0.652]	0.9382	0.7915	0.9083	HAYIR
R0_baseline/balbag	3427	0.269	0.4118	0.5456	[0.421-0.629]	0.9395	0.759	0.9083	HAYIR
R0_baseline/lgbm	3427	0.269	0.4081	0.5319	[0.399-0.625]	0.9308	0.7883	0.9083	HAYIR
R3_REVERSE_8020/balbag	1150	0.8	0.3843	0.5244	[0.368-0.609]	0.9317	0.7557	0.9083	HAYIR
R2_REVERSE_7030/lgbm	1318	0.7	0.3795	0.5238	[0.344-0.622]	0.9342	0.7394	0.9083	HAYIR
R2_REVERSE_7030/balbag	1318	0.7	0.345	0.5218	[0.399-0.619]	0.9369	0.6775	0.9083	HAYIR
R3_REVERSE_8020/lgbm	1150	0.8	0.298	0.5132	[0.209-0.724]	0.972	0.4528	0.9083	HAYIR

CFTR - Reverse-Distribution Sonuclari

COMBINED havuzu = MASTER + KANSER + PAH (CFTR panelin kendisi egitime girmiyor, LOO-CV ile test ediliyor). UYARI: CFTR n=21 benign -> tum karsilastirmalar dusuk-guven bandinda (CLAUDE.md 'CFTR DEGERLENDIRME GUVENI YOK').

Deney	n_train	benign_frac	LOO-MCC(prior)	Boot-F1(prior)	Boot-CI	Precision	Recall	FLOOR-F1	Floor Gecti mi
REF_S0c_PriorShift_NB20	-	-	-	0.8630		-	-	0.8955	HAYIR
R2_REVERSE_7030/lgbm	1377	0.7	0.5534	0.8377	[0.571-1.000]	1.0	0.7	0.8955	HAYIR
R1_REVERSE_6040/balbag	1606	0.6	0.6343	0.8108	[0.537-0.909]	0.9865	0.8111	0.8955	HAYIR
R3_REVERSE_8020/balbag	1200	0.8	0.5239	0.8053	[0.571-1.000]	1.0	0.6667	0.8955	HAYIR
R1_REVERSE_6040/lgbm	1606	0.6	0.5956	0.7809	[0.500-0.909]	0.9859	0.7778	0.8955	HAYIR
R2_REVERSE_7030/balbag	1377	0.7	0.5835	0.7792	[0.500-0.909]	0.9857	0.7667	0.8955	HAYIR
R0_baseline/lgbm	3685	0.262	0.5989	0.7407	[0.600-0.833]	0.9733	0.8111	0.8955	HAYIR
R0_baseline/balbag	3685	0.262	0.6128	0.7384	[0.479-0.833]	0.9737	0.8222	0.8955	HAYIR
R3_REVERSE_8020/lgbm	1200	0.8	0.4327	0.6729	[0.500-0.909]	0.9818	0.6	0.8955	HAYIR

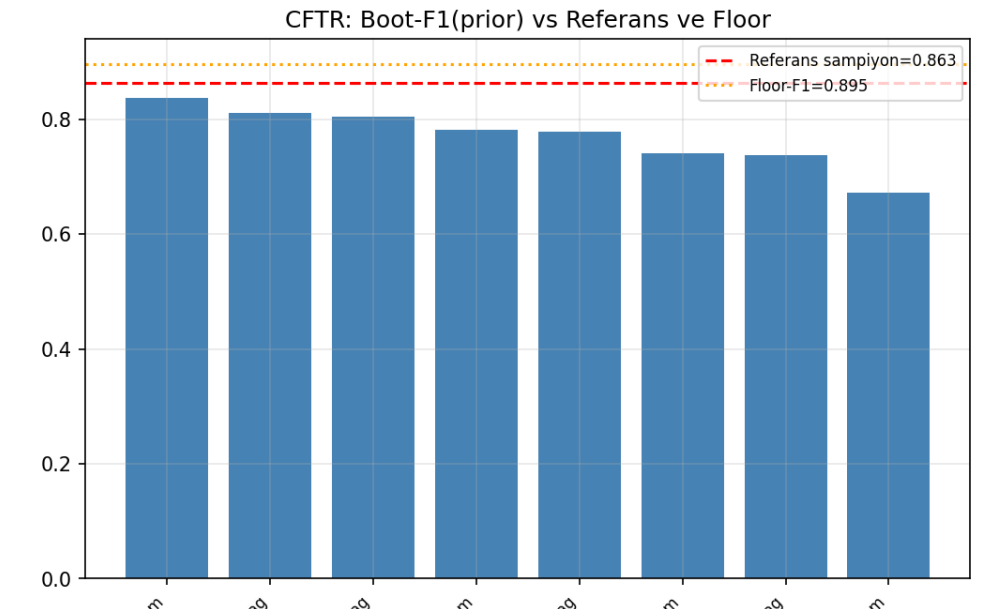
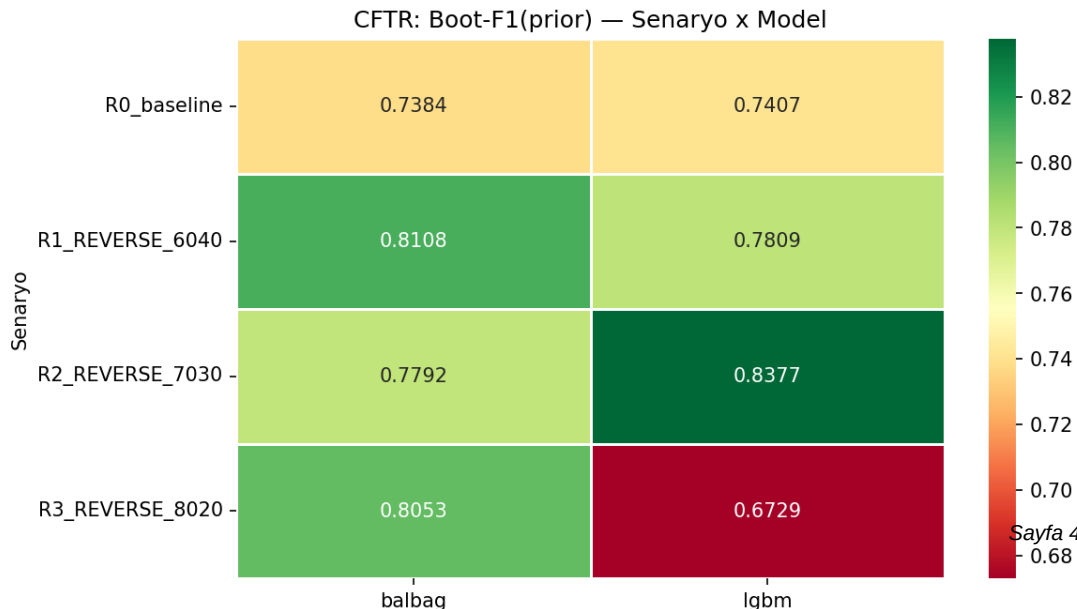
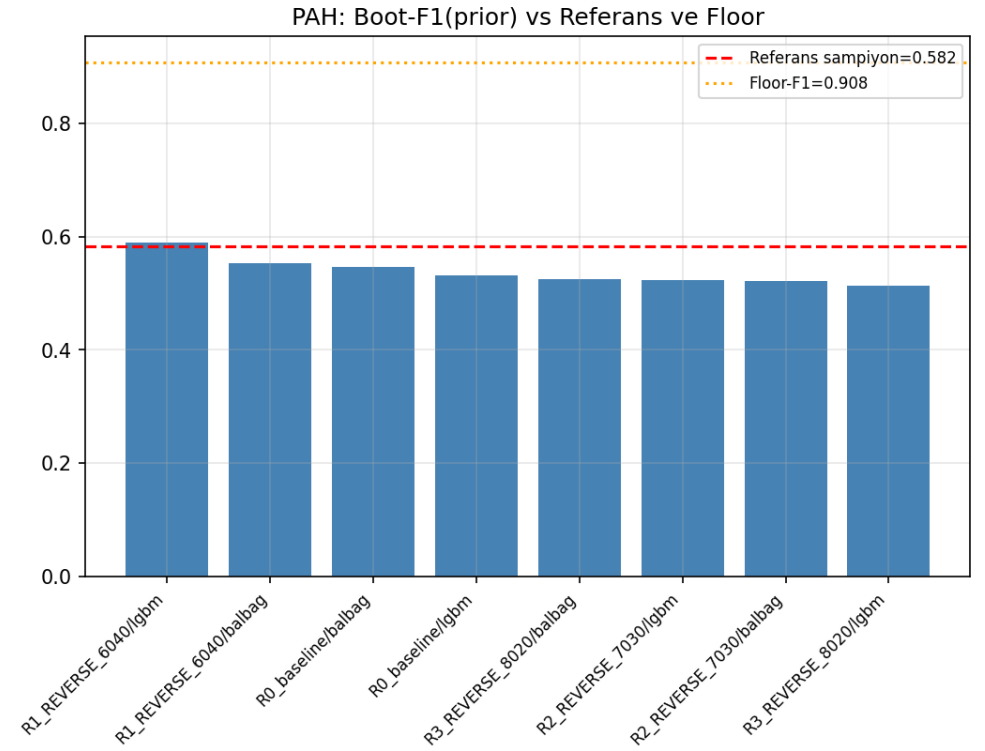
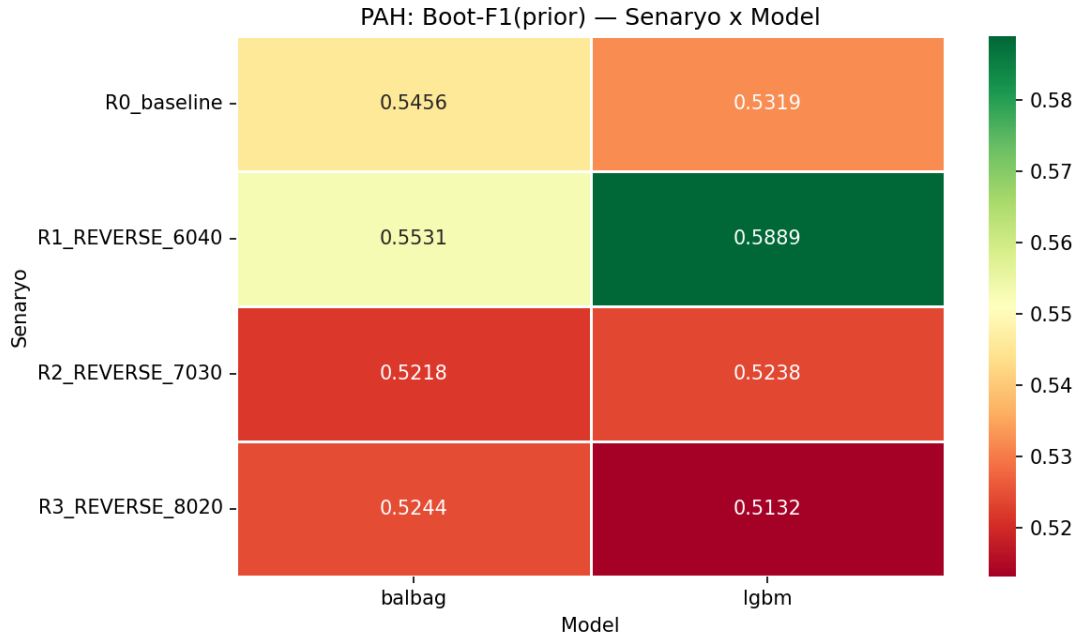
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Gorsellestirme

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Sonuc ve Karar

PAH: reverse-distribution KAZANDI. Delta=+0.0069.

CFTR: reverse-distribution NOTR/ZARARLI. Delta=-0.0253. (n=21 benign nedeniyle bu deltayi tek basina karar verici kabul etme -- CLAUDE.md'nin CFTR guven uyarisi burada da gecerli.)

Literatur baglantisi: Bu deney, reports/literature_research_panel_improvements_2026-07-24.md Bolum 4, Deney #2'yi tamamlar ('Reverse-distribution'i PAH ve CFTR'ye yay'). Sonuc, yol haritasindaki bir sonraki maddeye (missing-handling panel-bazli yeniden yargilama veya TabPFN-2.5) gecis icin veri saglar.